Aaryan S. Maralihalli

AI Backend Developer (Python | LLMs | Vector Search | Distributed Systems)

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Summary

Backend engineer building **AI-native, event-driven systems end-to-end** — **FastAPI** services, **LLM integration + RAG**, **vector & semantic search** (pgvector / FAISS), and high-volume **media pipelines** on object storage. Built a distributed, Kafka-backed platform that ingests audio/PDF media and serves ML over versioned HTTP contracts. Strong CS foundation (system design, DSA — 150+ LeetCode) and an ownership mindset: I ship what I design. CS graduate (Jul 2026), **IEEE-published**.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026
Enterprise AI identity-verification platform

- Integrated **production REST APIs / SDKs** for **6 enterprise fintech clients**; diagnosed user-journey drop-offs and shipped flow optimizations — took **3 live at >95% conversion**.
- Built **"HV Deals Bot"** — a **Python + Slack** backend service adopted org-wide for process visibility.

Integrated Genetic Solutions (IGS), Bangalore
Project Intern Jun 2024 – Aug 2024

- Supported **React + Django** integration for secure uploads and API telemetry on a HIPAA-compliant medical-AI product (**5M+ records**).

Selected Projects

StudyLoop — event-driven AI platform

- **FastAPI** microservices on **Kafka (Redpanda)** — gateway + worker services with **consumer groups, retries+backoff, DLQ + replay, idempotency, graceful degradation**. Ingests **audio + PDF media** to **object storage (MinIO/S3)** then ML serving (STT, embeddings, scoring) over **versioned HTTP contracts**, with **PostgreSQL + pgvector vector search (HNSW)** for **semantic retrieval** + LLM coverage scoring. Greenfield, solo-architected and built.

LANGTRON — AI memory assistant github.com/Aaryansm11/LANGTRON

- **Fine-tuned Mistral-7B (LoRA)**; dual-model (memory + reasoning) architecture; **FAISS semantic retrieval at sub-100ms** — LLM integration + retrieval orchestration.

SDIS — secure RAG service github.com/Aaryansm11/SDIS

- **FastAPI** retrieval platform with **JWT/RBAC**, PII redaction (spaCy), **FAISS embedding search over PostgreSQL**; Dockerized, **pytest + GitHub Actions CI**.

Langtron-KGS github.com/Aaryansm11/Langtron-KGS

- Documents to a **Neo4j knowledge graph** (spaCy NER + BERT relation extraction) with a **full-text / metadata search** REST API.

Self-Supervised Cardiac Model (IEEE-published) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- Contrastive representation learning (SimCLR) on MIMIC-III; **ROC-AUC 0.757**.

Skills

- **AI / LLM:** LLM integration, **RAG**, LoRA fine-tuning, embeddings, agent / multi-service orchestration, human-in-the-loop scoring, PyTorch, HuggingFace, LangChain, spaCy, BERT
- **Search: vector search (pgvector, FAISS)**, semantic retrieval, PostgreSQL full-text / metadata search, knowledge graphs (Neo4j)
- **Backend & APIs:** Python, FastAPI (async), Flask, REST API design, JWT/RBAC, SQLAlchemy/Alembic
- **Distributed / Infra:** Kafka (Redpanda), event-driven architecture, DLQ/retries/idempotency, Docker, AWS (EC2/S3/IAM/ object storage (MinIO/S3, multipart uploads), nginx, GitHub Actions CI
- **Data:** PostgreSQL (+ pgvector), Redis, Neo4j, MySQL — schema design & query optimization
- **Foundations:** system design, OOP, Data Structures & Algorithms (150+ LeetCode), DBMS

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026
- *Certifications:* WebGL (PESU I/O); Problem Solving — Intermediate (HackerRank); CITI Program